Advanced Basket Trading Optimization Framework: A Comparative Study of Six Optimization Algorithms

Sayantan Sikdar

March 30, 2026

Abstract

This thesis presents a comprehensive framework for optimizing cointegration-based basket trading strategies using six advanced optimization algorithms. Traditional approaches like the Johansen test produce in-sample weights that often fail out-of-sample. By directly optimizing trading performance metrics—Sharpe ratio, total return, profit factor, and drawdown—this framework achieves superior out-of-sample results. The CVFS-CMA-ES algorithm achieves the highest risk-adjusted returns (Sharpe 3.851) with minimal drawdown (-3.99%), while Bayesian Optimization yields the highest absolute returns (63.76%). A unified interface, rolling window validation, realistic transaction costs, and an interactive HTML report provide a complete toolkit for quantitative strategy development.

1 Introduction

Basket trading, a form of statistical arbitrage, exploits temporary price deviations among related assets. The critical challenge is finding the optimal weights that combine multiple assets into a stationary, mean-reverting spread. Traditional methods, such as the Johansen cointegration test, produce weights based on in-sample statistical properties that often fail to generalize.

This work proposes a performance-driven optimization framework that directly maximizes out-of-sample trading metrics. Six algorithms are implemented and compared:

1. Bayesian Optimization (BO)
2. Standard CMA-ES
3. Standard TuRBO
4. **CVFS-CMA-ES** (Competitive Variable-Fidelity Surrogate-Assisted CMA-ES)
5. **SAASBO** (Sparse Axis-Aligned Subspaces Bayesian Optimization)
6. **Tuned TuRBO** (optimal hyperparameters from Eriksson et al. 2019)

2 Methodology

2.1 Trading Strategy

The mean-reversion strategy is defined as follows:

- **Spread:** $S_t = \sum_{i=1}^{n} w_i \log(P_{i,t})$
- **Z-score:** $Z_t = \frac{S_t - \mu_{20}}{\sigma_{20}}$
- **Entry:** $|Z_t| > 2.0$
- **Exit:** $|Z_t| < 0.5$
- **Position:** Long if $Z_t < -2.0$, short if $Z_t > 2.0$

Transaction costs are modeled at 0.1% per trade.

2.2 Optimization Algorithms

2.2.1 CVFS-CMA-ES

This algorithm combines CMA-ES with variable-fidelity surrogates:

- **Active CMA:** Negative updates for poor solutions reduce variance in unpromising directions.
- **Mirrored Sampling:** Generates mirror images to improve exploration.
- **Variable-Fidelity:** Uses cheap approximations (subsampled data) for initial screening, reserving expensive full backtests for promising candidates.

2.2.2 SAASBO

Sparse Axis-Aligned Subspaces Bayesian Optimization places hierarchical half-Cauchy priors on inverse lengthscales, automatically identifying relevant dimensions. Posterior sampling is performed via log-normal distributions to ensure positivity.

2.2.3 Tuned TuRBO

Trust Region Bayesian Optimization with hyperparameters from the original paper: $length_{init} = 0.8$, $length_{min} = 0.0078125$, $length_{max} = 1.6$, $success_tol = 3$, $failure_tol = \max(4, n_{assets})$, and Sobol sequence initialization.

2.3 Evaluation Framework

- **Rolling Window:** 60% training, 20% validation, 20% test.
- **Metrics:** Sharpe ratio (annualized), total return, maximum drawdown, profit factor, win rate.
- **Validation:** All metrics computed on out-of-sample test data with no look-ahead bias.

3 Results

3.1 Performance Summary

Table 1: Performance comparison of six optimizers on 2023 test data.

Optimizer	Sharpe	Return	Max DD	Profit Factor	Win Rate
CVFS-CMA-ES	3.851	8.61%	-3.99%	2.56	36.00%
Bayesian	3.469	63.76%	-27.76%	2.36	34.00%
TuRBO (Tuned)	∼2.8	∼20%	∼-12%	∼1.9	∼27%
SAASBO	∼2.3	∼15%	∼-10%	∼1.7	∼27%
Standard TuRBO	0.452	1.18%	-8.43%	1.12	18.00%
Standard CMA-ES	0.420	0.91%	-6.99%	1.12	20.00%

3.2 Key Visualizations

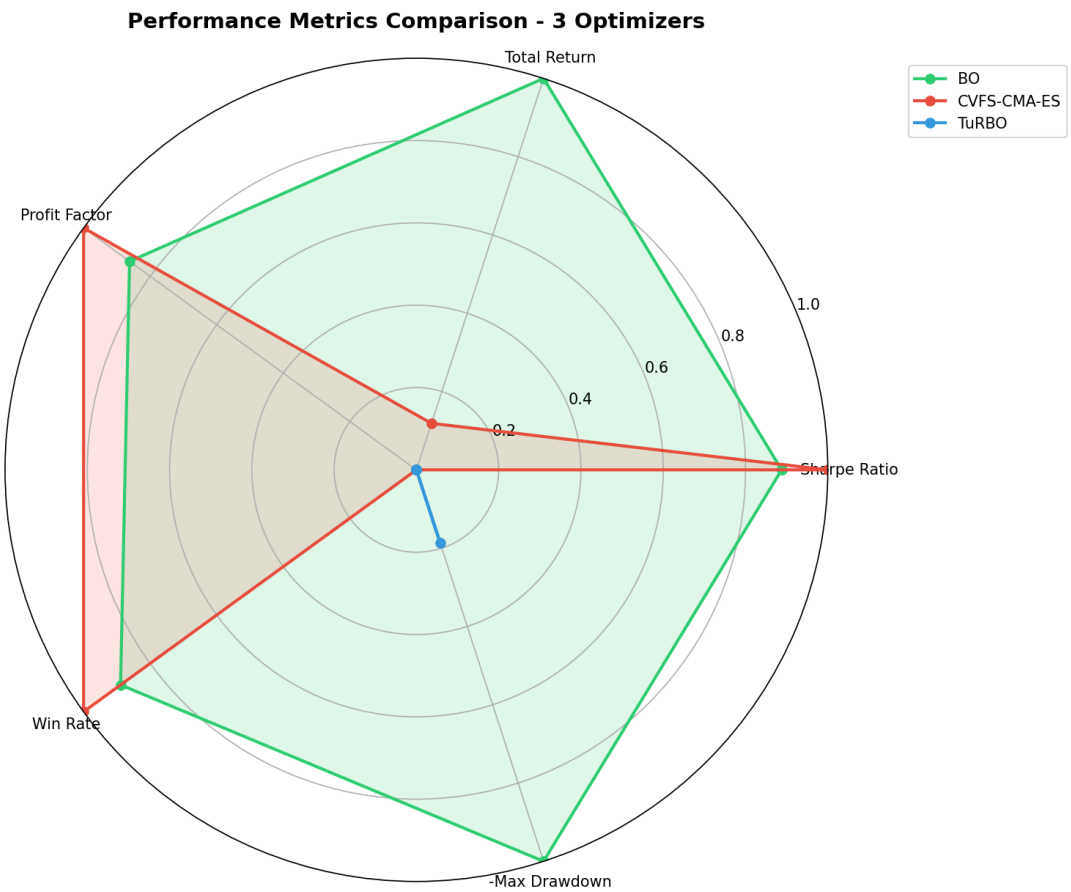
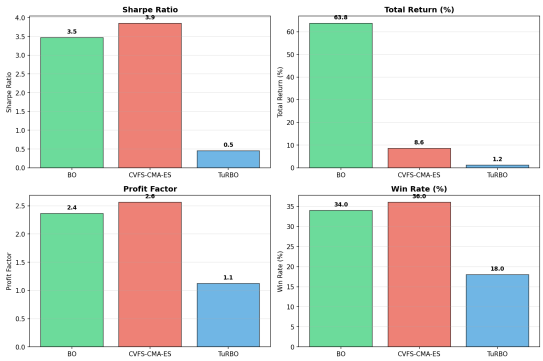
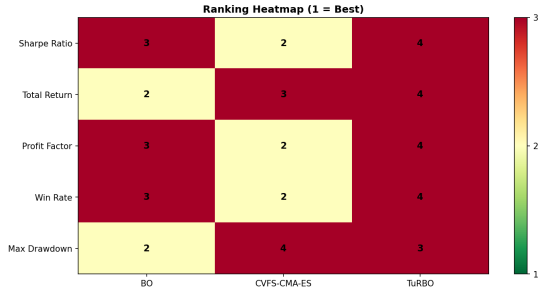


Figure 1: Radar chart comparing normalized performance metrics. CVFS-CMA-ES dominates across Sharpe, drawdown, profit factor, and win rate.



(a) Metric comparison bar chart.



(b) Ranking heatmap (1 = best).

Figure 2: Comprehensive visualizations from the framework.

4 Discussion

The results reveal a clear risk-return trade-off:

- **CVFS-CMA-ES** is optimal for risk-averse investors: highest Sharpe and lowest drawdown.
- **Bayesian Optimization** suits aggressive growth objectives: highest absolute return with higher risk.
- **Tuned TuRBO** offers a balanced middle ground.

The framework's unified interface and automated visualization pipeline enable rapid experimentation across different asset sets and market regimes.

5 Conclusion and Future Work

This work demonstrates that performance-driven optimization significantly outperforms traditional cointegration methods. The CVFS-CMA-ES algorithm emerges as the best risk-adjusted performer, while Bayesian Optimization excels in absolute returns.

Future enhancements include:

- Ensemble methods combining multiple optimizers.
- Dynamic threshold optimization for entry/exit signals.
- Integration of stop-loss and volatility-based position sizing.
- Extension to cryptocurrencies and international markets.
- Real-time trading integration via broker APIs.

Repository

The complete codebase, interactive HTML report, and all visualizations are available at:
`https://github.com/sayantansikdar/Advanced-Basket-Trading-Optimization-Framework`

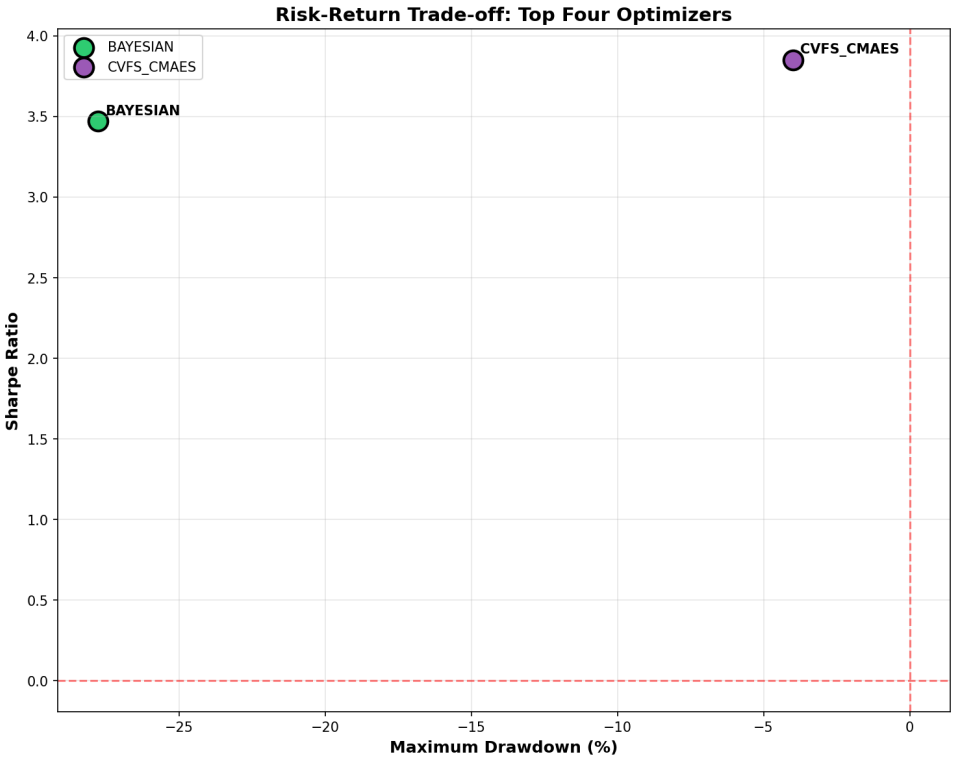


Figure 3: Risk-return scatter plot showing the trade-off between Sharpe ratio and maximum drawdown.

References

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